

Bulgarian Solitaire: Recurrent Necklaces, Every Fixed Iterate, and the Full Finite Koopman Spectrum

Anonymous structural certificate HCS-C190

Abstract

For Bulgarian solitaire on every integer-partition set $\mathcal{P}(N)$, we turn Brandt's attributed recurrent classification into a complete periodic ledger. The unique decomposition $N = \binom{k}{2} + r$, $0 \leq r < k$, identifies recurrent partitions with length- k , weight- r binary words, and one solitaire move becomes rotation. We derive a closed formula for every positive-iterate fixed count, then recover all least periods, primitive cycles, and the finite Artin–Mazur zeta by Möbius inversion. The finite cycle factors also give the full Koopman algebraic spectrum, including the transient zero multiplicity. The finite census is a regression oracle, not the proof of the all- N theorem.

Keywords: Bulgarian solitaire; integer partitions; binary necklaces; primitive cycles; dynamical zeta; Koopman spectrum.

中文摘要

本文研究任意整数分拆集合上的保加利亚单人纸牌动力学。利用归属于 Brandt 的循环分拆分类，将循环状态与长度为 k 、重量为 r 的二进制词对应，其中 $N = \binom{k}{2} + r$ 且 $0 \leq r < k$ ；一次动力学迭代正好对应循环旋转。由此得到所有正迭代固定点数的闭式公式，再用 Möbius 反演恢复全部最小周期、原始循环、有限动力学 ζ 函数与包含瞬态零重数的完整有限 Koopman 代数谱。有限枚举仅用于回归检验，不替代全参数定理。

关键词：保加利亚单人纸牌；整数分拆；二进制项链；原始循环；动力学 ζ 函数；Koopman 谱。

1 The noninvertible map and Brandt coordinates

Write a partition of N as $\lambda = (\lambda_1 \geq \dots \geq \lambda_m > 0)$. The Bulgarian move is

$$T_N(\lambda) = \text{sort}(m, \lambda_1 - 1, \dots, \lambda_m - 1),$$

after zero parts are deleted. This map is generally noninjective. Every $N \geq 1$ has a unique representation

$$N = \binom{k}{2} + r, \quad k \geq 2, \quad 0 \leq r < k.$$

Brandt's cyclic-partition theorem [1], treated also by Akin and Davis [2], says that the recurrent set consists exactly of

$$\phi(w) = \text{positive parts of } (k-1, k-2, \dots, 0) + w, \quad w \in \{0, 1\}^k, \quad |w| = r.$$

For the right rotation $(\rho w)_i = w_{i-1 \bmod k}$, direct subtraction of the first column gives $T_N \phi = \phi \rho$. States correspond to words; dynamical cycles correspond to binary necklaces.

2 Every fixed iterate and every primitive cycle

Theorem 1 *Let $t \geq 1$ and $g = \gcd(k, t)$. Then*

$$F_t := \# \text{Fix}(T_N^t) = \begin{cases} \binom{g}{rg/k}, & (k/g) \mid r, \\ 0, & \text{otherwise.} \end{cases}$$

For each $d \mid k$, the number P_d of points of least period d and the number C_d of d -cycles are

$$P_d = \sum_{e \mid d} \mu(d/e) F_e, \quad C_d = P_d/d.$$

Consequently the zeta function of the full noninvertible finite map is

$$\zeta_{T_N}(z) = \exp \left(\sum_{t \geq 1} \frac{F_t}{t} z^t \right) = \prod_{d \mid k} (1 - z^d)^{-C_d}. \quad (1)$$

Proof. The index rotation ρ^t has g cycles, each of length k/g . A fixed binary word is constant on each index cycle, so its weight is a multiple of k/g ; choosing the rg/k one-cycles gives the displayed binomial. A point fixed by T_N^t is periodic and hence recurrent, so no transient partition adds to this count. Divisor Möbius inversion gives P_d , and division by d gives C_d . Substituting $F_t = \sum_{d \mid t} d C_d$ into the exponential definition proves (1). $\square$

3 The full finite Koopman algebraic spectrum

Let $p(N) = |\mathcal{P}(N)|$ and let $(U_N f)(\lambda) = f(T_N \lambda)$ on all complex functions on $\mathcal{P}(N)$. Ordering each functional-graph component from its transient trees toward its cycle gives

$$\det(\xi I - U_N) = \xi^{p(N) - \binom{k}{r}} \prod_{d|k} (\xi^d - 1)^{C_d}, \quad \det(I - zU_N) = \prod_{d|k} (1 - z^d)^{C_d} = \zeta_{T_N}(z)^{-1}. \quad (2)$$

Thus zero has algebraic multiplicity $p(N) - \binom{k}{r}$. If $\omega_k = e^{2\pi i/k}$, the nonzero multiplicities are

$$\text{mult}(\omega_k^j) = \sum_{\substack{d|k \\ k|jd}} C_d, \quad \text{Tr}(U_N^t) = F_t.$$

Indeed, transient vertices contribute only a zero diagonal block to the characteristic polynomial, whereas a d -cycle contributes $\xi^d - 1$ and every d th root once. Formula (2) determines eigenvalues with algebraic multiplicities, not the nilpotent Jordan block sizes in the transient sector.

For $N = 8 = \binom{4}{2} + 2$, the six recurrent words form one 2-cycle and one 4-cycle. Hence $(F_1, F_2, F_3, F_4) = (0, 2, 0, 6)$, $\zeta_{T_8}(z) = ((1 - z^2)(1 - z^4))^{-1}$, zero has multiplicity $22 - 6 = 16$, and the four fourth-root multiplicities are $(2, 1, 2, 1)$. The sentinel is obtained both from all 22 partitions of 8 and from the word model.

4 Reflection reversal on the recurrent core

Use indices in $\mathbb{Z}/k\mathbb{Z}$ and define $(Qw)_i = w_{-i}$. Then $Q^2 = 1$ and a direct index calculation gives

$$Q\rho Q = \rho^{-1}.$$

Transporting Q through ϕ produces an involutory reversor of the recurrent partition permutation. More generally, each $\rho^a Q$ is an involutory reversor, giving k phase-labelled reflection formulas; these formulas need not act distinctly when the weight layer is nonfaithful. The corresponding recurrent Koopman permutation is unitary; reflection followed by coefficientwise conjugation is an antiunitary reversor. No such inverse relation is claimed on all of $\mathcal{P}(N)$ because T_N is generally noninjective.

5 Triangular boundary and evidence scope

If $r = 0$, the sole word is 0^k and the sole recurrent partition is the staircase $(k - 1, k - 2, \dots, 1)$. It is fixed, so $F_t = 1$ and $\zeta_{T_N}(z) = (1 - z)^{-1}$; the recurrent Koopman spectrum is $\{1\}$, while the full operator has zero algebraic multiplicity $p(N) - 1$. The conclusion concerns the recurrent core; triangular deck size does not erase the transient basin in general.

The executable census covers every $1 \leq N \leq 40$. A producer enumerates 757 recurrent words and their 114 cycles. A separate checker constructs all 215,307 integer partitions, follows the actual noninvertible map, and recovers exactly the same recurrent states and cycles. This finite agreement tests implementations and conventions; it does not establish Brandt's all- N classification or classify complete transient trees and hitting times.

The independent checker passes 658,664 assertions. A separate SymPy path passes 2,210 checks, producer replay is byte exact, and the hostile suite rejects 118 semantic mutations after their payload hashes are repaired plus one stale-hash attack. These are implementation checks, not an independent mathematical review.

6 Route-A stop and limitations

Periodic cycles and their finite determinant are exact, but partition and necklace data contain no intrinsic rational-prime carrier, prime-power weight, or logarithmic prime clock. No target divisor, functional equation, continuation, counting law, or Weil compression follows. The finite unitary on the recurrent core is therefore a formal operator hint only:

$$(A0, A1, A2, A3, A4) = (\text{FAIL}, \text{WEAK}, \text{FAIL}, \text{FAIL}, \text{FORMAL HINT}).$$

The overall verdict is `ROUTE_A_REJECTED`; Route B is false. We do not claim a complete transient tree, an exact hitting-time distribution, nilpotent Jordan block sizes, a global reversor, or priority for the Brandt and Akin–Davis inputs.

Scope. No target zero or prime table, arithmetic local datum, Euler factor, root number, automorphy input, target divisor, or Hilbert–Polya operator is used. The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript. **Ethics.** No human, animal, clinical, personal, or sensitive data are used; approval is not applicable. **Author contributions (CRediT).** This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, and Writing; AI systems are not authors. **Funding and conflicts.** No external funding is reported; no conflict is known. **AI-use disclosure.** An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

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